

HVAC Opta Parc

Manufacturing & Wiring Guide

Configs A · B · C · D — 1P + start cap MCSA

Project: hvac-opta-parc
Firmware: MooreviewOptaMqttSt 2.3.81
Generated: 2026-08-14

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Configuration matrix

Cfg	HVAC_CFG	Description	Hardware	MCSA wiring	Expansion	Opta
A	0	Single air handler	Base	1P + start cap (blower)	—	1× Opta
B	1	Single condenser	Base	1P + start cap (compressor + fan)	—	1× Opta
C	2	Dual air handler	Opta + D1608E (digital expansion)	1P + start cap (each blower)	D1608E	1× Opta
D	3	Dual condenser	2× Opta MCSA + D1608E on primary	1P + start cap (compressor + fan per unit)	D1608E	2× Opta

MCSA — single-phase with start capacitor (all fan & compressor configs)

- 1P + start cap (MV_MOTOR_WIRE_1P_CAP) — not plain 1P (no cap).
- M4 edge AI model: single-phase-start-v1.
- Run CT on line/run terminal; start-winding CT on start cap branch. End-of-start = start CT current drops.
- Config A: dedicated run CT (I2) + start-winding CT (I3) on blower.
- Configs B/D: HVAC layout — run CT on I1 (compressor) and I2 (fan); start-cap diagnostics via MCSA on run CT.

Shared hardware conventions

- Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA)
- 24 VDC field supply — leak DI pull-ups and NTC dividers
- 0–1 V DC CT on run and start-winding branches (1P+cap motors)
- 10K NTC thermistors — Opta 24 VDC divider
- Spot leak detectors — dry contact to +24 V

Shared wiring conventions

- All monitored fans and compressors: single-phase with start capacitor (1P+cap).
- 1P + start cap (MV_MOTOR_WIRE_1P_CAP) · Edge model single-phase-start-v1.
- Digital inputs: +24 VDC active (contact closed = ON).
- CT inputs: 0–10 V ADC; use 0–1 V CT transmitters on run and start CTs.
- NTC: 24 V !' 10K NTC !' analog in !' 10K to GND.
- Set HVAC_CFG tag (0–3) before field test.

WiFi AP, /setup & MQTT Parc (all configs)

- Enable WiFi AP on <http://<opta-ip>/setup> or <http://192.168.4.1:8080/setup>
- Default AP SSID: MooreVIEW-Opta (password "e 8 chars)
- Pages: /setup, /mcsa, /api/status
- MQTT test !' mooreview/v1/{deviceId}/setup-test
- MQTT password "d 47 characters

Reference: docs/OPTA_PARC_CLOUD.md · scripts/hvac-opta-parc/README.md

Config summary — I/O counts

Cfg	Motors (1P+cap)	Run CT	Start CT	Leak DI	NTC	Expansion
A	1 blower	I2	I3	1	2	None
B	comp + fan	I1, I2	via MCSA	—	2	None
C	2 blowers	I2, I3	—	2	4	D1608E
D	2× comp + fan	I1, I2 ×2	via MCSA	2 DI	4	D1608E + 2nd Opta

Config A — Single air handler

HVAC_CFG = 0 · hvac-opta-parc — Config A (HVAC_CFG=0)

MooreVIEW project: hvac-opta-parc · ST: logic/hvac_opta_parc.st

Device template: arduino_opta_parc

Device ID(s): hvac_opta_01

Hardware: Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA), Base Opta only — blower 1P+cap (run + start CT)

Bill of materials / hardware

- Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA)
- Base Opta only — blower 1P+cap (run + start CT)

Wiring notes

- All monitored fans and compressors: single-phase with start capacitor (1P+cap).
- 1P + start cap (MV_MOTOR_WIRE_1P_CAP) · Edge model single-phase-start-v1.
- Digital inputs: +24 VDC active (contact closed = ON).
- CT inputs: 0–10 V ADC; use 0–1 V CT transmitters on run and start CTs.
- NTC: 24 V !' 10K NTC !' analog in !' 10K to GND.
- Set HVAC_CFG tag (0–3) before field test.
- Blower run CT !' I2; start-winding CT !' I3 (same 1P+cap motor).
- I1 pan leak (digital). I6/I7 supply and return NTC.

Digital inputs

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
I1	DI	—	AHU1_PAN_LEAK	Pan spot leak

Blower — 1P + start cap CTs

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
Blower	1P+cap	I2	I3	AHU1_FAN_AMPS / AHU1_START_AMPS

Run CT I2 · st

Temperature (NTC)

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
I6	AI	—	AHU1_SUPPLY_TEMP_F	Supply air
I7	AI	—	AHU1_RETURN_TEMP_F	Return air

Relay outputs

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
R1	DO	—	R1	AHU alarm
R3	DO	—	R3	Site OK

Manufacturing test checklist

1. Flash firmware v2.3.81+; label device ID
2. Set HVAC_CFG=0
3. Wire blower run CT I2, start CT I3, leak I1, NTC I6/I7
4. Calibrate NTC; verify AHU1_FAN_AMPS and AHU1_START_AMPS
5. Test leak and CT overload alarms

Config B — Single condenser

HVAC_CFG = 1 · hvac-opta-parc — Config B (HVAC_CFG=1)

MooreVIEW project: hvac-opta-parc · ST: logic/hvac_opta_parc.st

Device template: arduino_opta_mcsa_monitor

Device ID(s): hvac_opta_01

Hardware: Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA),Base Opta — MCSA HVAC layout,Compressor + condenser fan: both 1P+cap

Bill of materials / hardware

- Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA)
- Base Opta — MCSA HVAC layout
- Compressor + condenser fan: both 1P+cap

Wiring notes

- All monitored fans and compressors: single-phase with start capacitor (1P+cap).
- 1P + start cap (MV_MOTOR_WIRE_1P_CAP) · Edge model single-phase-start-v1.
- Digital inputs: +24 VDC active (contact closed = ON).
- CT inputs: 0–10 V ADC; use 0–1 V CT transmitters on run and start CTs.
- NTC: 24 V !' 10K NTC !' analog in !' 10K to GND.
- Set HVAC_CFG tag (0–3) before field test.
- Open <http://<opta-ip>/mcsa> (or /hvac alias)
- Enable MCSA monitor; I/O layout = HVAC
- deviceType = 5 (all loads) or compressor + fan classes
- Set each motor wiring to 1P + start cap (not plain 1P)
- I1 = compressor run CT (COMP_AMPS, COMP_FLT)
- I2 = condenser fan run CT (FAN_AMPS, FAN_FLT)
- Start-cap health from run CT waveform + M4 edge AI (single-phase-start-v1)
- I3/I4 NTC = high / low side refrigerant temps

/mcsa setup (1P + start cap)

1. Open <http://<opta-ip>/mcsa> (or /hvac alias)
2. Enable MCSA monitor; I/O layout = HVAC
3. deviceType = 5 (all loads) or compressor + fan classes
4. Set each motor wiring to 1P + start cap (not plain 1P)
5. Compressor: assetId compressor, run CT on I1
6. Condenser fan: assetId fan, run CT on I2
7. M4 edge AI uses single-phase-start-v1 for start-cap health
8. Save !' Reboot; verify COMP_FLT / FAN_FLT and mcsa[] telemetry

MCSA motors — 1P + start cap (run CT)

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description	
Compressor	1P+cap	I1	—	COND1_COMP_AMPS	Run CT !' COM
Cond. fan	1P+cap	I2	—	COND1_FAN_AMPS	Run CT !' FAN

Thermistors

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
I3	NTC	—	COND1_HI_TEMP_F	High side T1_C
I4	NTC	—	COND1_LO_TEMP_F	Low side T2_C

/mcsa configuration

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
1	Setup	1P+cap	—	Open <a href="http://<opta-ip>/mcsa">http://<opta-ip>/mcsa (or /hvac alias)
2	Setup	1P+cap	—	Enable MCSA monitor; I/O layout = HVAC
3	Setup	1P+cap	—	deviceType = 5 (all loads) or compressor + fan classes
4	Setup	1P+cap	—	Set each motor wiring to 1P + start cap (not plain 1P)
5	Setup	1P+cap	—	Compressor: assetId compressor, run CT on I1
6	Setup	1P+cap	—	Condenser fan: assetId fan, run CT on I2
7	Setup	1P+cap	—	M4 edge AI uses single-phase-start-v1 for start-cap health
8	Setup	1P+cap	—	Save !' Reboot; verify COMP_FLT / FAN_FLT and mcsa[] telemetry

Manufacturing test checklist

1. Set HVAC_CFG=1; configure /mcsa per MCSA setup list
2. Wire comp run CT I1, fan run CT I2, NTC I3/I4
3. Verify COMP_FLT, FAN_FLT, T1_C, T2_C in telemetry
4. Confirm mcsa[] shows compressor and fan load groups

Config C — Dual air handler

HVAC_CFG = 2 · hvac-opta-parc — Config C (HVAC_CFG=2)

MooreVIEW project: hvac-opta-parc · ST: logic/hvac_opta_parc.st

Device template: arduino_opta_parc

Device ID(s): hvac_opta_01

Hardware: Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA), 24 VDC field supply — leak DI pull-ups and NTC dividers, D1608E (AFX00005) slot 1 — second leak DI

Bill of materials / hardware

- Arduino Opta WiFi (AFX00001) — flash MooreviewOptaMqttSt v2.3.81+ (M7 + M4 MCSA)
- 24 VDC field supply — leak DI pull-ups and NTC dividers
- D1608E (AFX00005) slot 1 — second leak DI

Wiring notes

- All monitored fans and compressors: single-phase with start capacitor (1P+cap).
- 1P + start cap (MV_MOTOR_WIRE_1P_CAP) · Edge model single-phase-start-v1.
- Digital inputs: +24 VDC active (contact closed = ON).
- CT inputs: 0–10 V ADC; use 0–1 V CT transmitters on run and start CTs.
- NTC: 24 V ! 10K NTC ! analog in ! 10K to GND.
- Set HVAC_CFG tag (0–3) before field test.
- Each blower 1P+cap — run CT only on I2 (AHU1) and I3 (AHU2) in this layout.
- AHU1 leak I1; AHU2 leak X1_I1 on D1608E.
- Four NTC: I6/I7 AHU1 supply/return; I8/I4 AHU2 supply/return.

Digital — leak

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
I1	DI	—	AHU1_PAN_LEAK	AHU 1 pan leak
X1_I1	DI	—	AHU2_PAN_LEAK	AHU 2 pan leak (D1608E)

Blowers — 1P + start cap (run CT each)

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description	
AHU 1 blower	1P+cap	I2	—	AHU1_FAN_AMPS	Run CT
AHU 2 blower	1P+cap	I3	—	AHU2_FAN_AMPS	Run CT

Temperature (NTC)

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
I6	AI	—	AHU1_SUPPLY_TEMP_F	AHU 1 supply
I7	AI	—	AHU1_RETURN_TEMP_F	AHU 1 return
I8	AI	—	AHU2_SUPPLY_TEMP_F	AHU 2 supply
I4	AI	—	AHU2_RETURN_TEMP_F	AHU 2 return

Manufacturing test checklist

1. Mount D1608E; scan on /setup
2. Set HVAC_CFG=2
3. Wire run CTs on I2 and I3 for 1P+cap blowers
4. Wire leaks and four NTCs; test each AHU alarm

Config D — Dual condenser

HVAC_CFG = 3 · hvac-opta-parc — Config D (HVAC_CFG=3)

MooreVIEW project: hvac-opta-parc · ST: logic/hvac_opta_parc.st

Device template: arduino_opta_mcsa_monitor

Device ID(s): hvac_opta_01, hvac_opta_02

Hardware: Primary + secondary Opta — each MCSA HVAC, Primary D1608E — condenser 2 lockout DI, Each unit: compressor + fan 1P+cap, 0–1 V DC CT on run and start-winding branches (1P+cap motors), 10K NTC thermistors — Opta 24 VDC divider, Spot leak detectors — dry contact to +24 V

Bill of materials / hardware

- Primary + secondary Opta — each MCSA HVAC
- Primary D1608E — condenser 2 lockout DI
- Each unit: compressor + fan 1P+cap
- 0–1 V DC CT on run and start-winding branches (1P+cap motors)
- 10K NTC thermistors — Opta 24 VDC divider
- Spot leak detectors — dry contact to +24 V

Wiring notes

- All monitored fans and compressors: single-phase with start capacitor (1P+cap).
- 1P + start cap (MV_MOTOR_WIRE_1P_CAP) · Edge model single-phase-start-v1.
- Digital inputs: +24 VDC active (contact closed = ON).
- CT inputs: 0–10 V ADC; use 0–1 V CT transmitters on run and start CTs.
- NTC: 24 V ! 10K NTC ! analog in ! 10K to GND.
- Set HVAC_CFG tag (0–3) before field test.
- Repeat MCSA HVAC wiring on each Opta (unique device ID).
- Per unit: I1 comp run CT, I2 fan run CT, I3/I4 hi/lo NTC.
- Primary X1_I1/X1_I2 = condenser 2 lockout / shutdown.

/mcsa setup (1P + start cap)

1. Open <http://<opta-ip>/mcsa> (or /hvac alias)
2. Enable MCSA monitor; I/O layout = HVAC
3. deviceType = 5 (all loads) or compressor + fan classes
4. Set each motor wiring to 1P + start cap (not plain 1P)
5. Compressor: assetId compressor, run CT on I1
6. Condenser fan: assetId fan, run CT on I2
7. M4 edge AI uses single-phase-start-v1 for start-cap health
8. Save ! Reboot; verify COMP_FLT / FAN_FLT and mcsa[] telemetry

Condenser 1 — primary Opta (1P+cap motors)

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description	
Compressor	1P+cap	I1	—	COND1_COMP_AMPS	Run CT
Cond. fan	1P+cap	I2	—	COND1_FAN_AMPS	Run CT
I3	NTC	—	COND1_HI_TEMP_F	High side	

I4	NTC	—	COND1_LO_TEMP_F	Low side
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Condenser 2 — secondary Opta (1P+cap motors)

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description	
Compressor	1P+cap	I1	—	COND2_COMP_AMPS	Run CT
Cond. fan	1P+cap	I2	—	COND2_FAN_AMPS	Run CT
I3	NTC	—	COND2_HI_TEMP_F	High side	
I4	NTC	—	COND2_LO_TEMP_F	Low side	

Primary D1608E — interlocks

Terminal	Point	MCSA / wiring	MooreVIEW tag	Description
X1_I1	DI	—	COND2_LOCKOUT	Condenser 2 lockout
X1_I2	DI	—	COND2_SHUTDOWN	Optional shutdown

Manufacturing test checklist

1. Flash both Optas; unique device IDs
2. Set HVAC_CFG=3; /mcsa 1P+cap on both units
3. Enable opta_hvac_02 driver in MooreVIEW
4. Verify COND1_* and COND2_* MCSA faults independently

MooreVIEW project files

- Project: data/projects/hvac-opta-parc.est.json
- Generate: node scripts/hvac-opta-parc/generate-est.js
- Tags fixture: st/fixtures/tags.hvac_opta_parc.json
- ST program: st/logic/hvac_opta_parc.st
- Starter manifest: st/fixtures/hvac_opta_parc_starter.json

Revision

Document generated 2026-08-14 from scripts/hvac-opta-parc/build-hvac-mfg-pdf.js. Firmware baseline: firmware/arduino-opta-mqtt-st/MooreviewOptaMqttSt v2.3.63+.